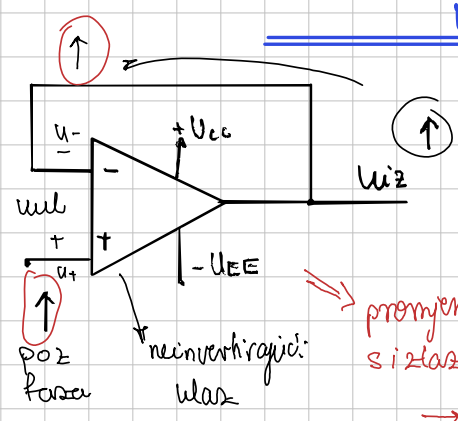
POJAČALA S POVRATNOM VEZOM



$$u_{iz} = A_{vop} \cdot \underbrace{(u_+ - u_-)}_{\text{dif. signal}}$$

promjena je u istom smjeru – kada vratimo signal s izlaza na ulaz

→ na ulazu smanjuje dif. signal

→ $U_d \downarrow$ = negativna povratna veza

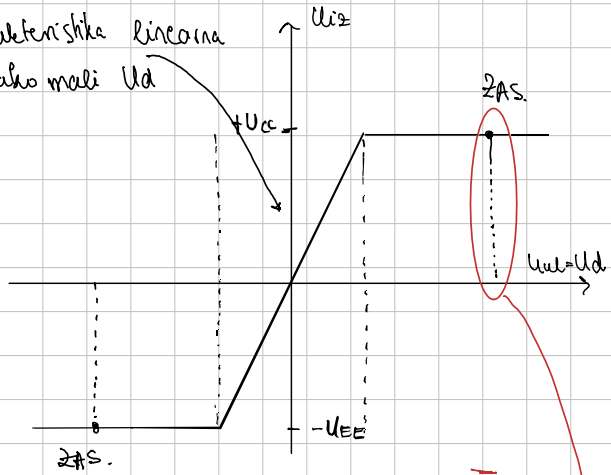
$A_{vop} \to \infty$

$$\frac{u_{iz}}{\infty} = u_+ - u_- \to \boxed{u_+ = u_-}$$

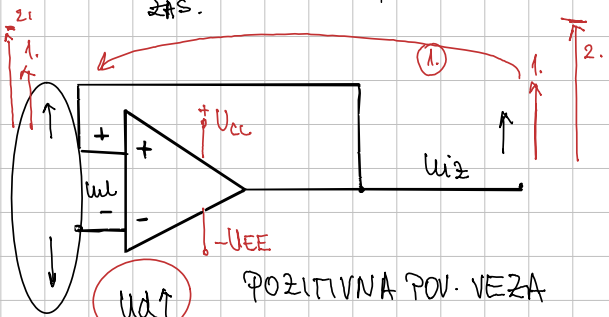
virtualni kratki spoj

Karakteristika linearna

→ jako mali U_d



Lin. pojačala: INVERTIRAJUĆE
NE-INVERT.
INSRUM. POJ.



POZITIVNA POV. VEZA

$U_d \uparrow$

→ po iznosu raste

– zarišćavamo na poz. liniji zasićenja

POZ. POV. VEZA

- Komparator
- Oscilator

Povratna veza

→ info ili signal vraća s izlaza na ulaz gdje uspoređuje s ulaznom veličinom

· negativna: smanjuje izlazni signal

$A_v = -\frac{R_2}{R_1}$ ... inv. pojačalo

$A_v = \left(1+\frac{R_2}{R_1}\right)$ .... neinv. pojačalo

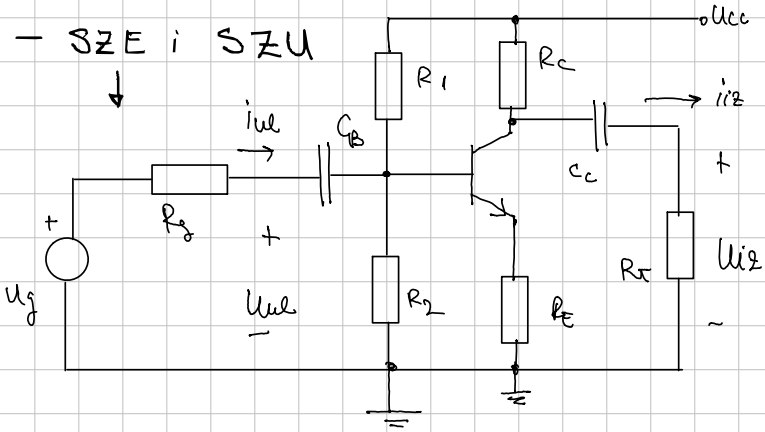
} stabilizira pojačanje

↳ smanjuje nelinearna izobličenja i povećava širinu frekv. pojasa
⟹ približno idealna

· pozitivna: povećava izlazni signal

Primjer: pojačalo s povratnom vezom

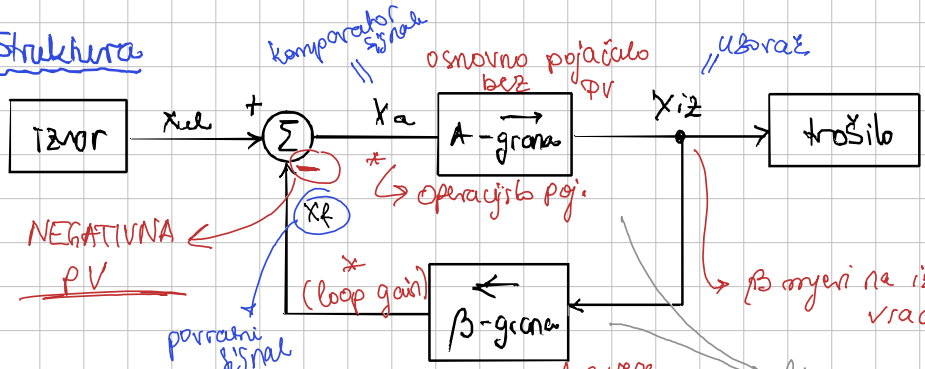
– SZE i SZU



→ preko R_E dio izlaznog signala vraća se na ulaz

$$A_v \approx \frac{R_C \parallel R_T}{R_E}$$

Struktura



komparator signala

osnovno pojačalo bez PV

uzorak

* → operacijsko poj.

NEGATIVNA PV

* (loop gain)

povratni signal

β mjeri na izlazu i vraća to na ulaz

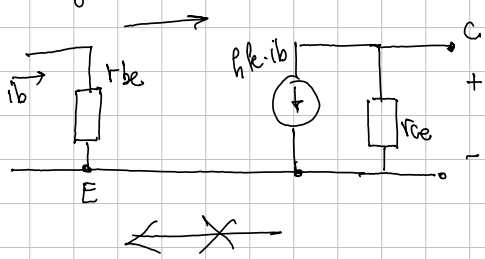
grana povratne veze

* koeficijent PV

linearne komponente (vrlo bitno za A granu) zbog nadomještanja

* signal teče u samo jednom smjeru

↳ npr



$X_{iz} = X_a \cdot A$

$X_{iz} = A(X_{ul} - X_f)$

$X_{iz} = A \cdot (X_{ul} - \beta X_{iz})$

$X_{iz} = A X_{ul} - A\beta X_{iz}$

$$\Rightarrow \boxed{\frac{X_{iz}}{X_{ul}} = A_f = \frac{A}{1+\beta A}}$$

pojačanje uz P.V.

faktor PV

Stabilizacija pojačanja

$$A_f = \frac{A}{1+\beta A}$$

$$A_f = \frac{A}{1+\beta A} \rightarrow dA_f = \frac{dA_f}{dA} \cdot dA = \frac{1(1+\beta A) - A \cdot \beta}{(1+\beta A)^2} dA$$

$$dA_f = \frac{1}{(1+\beta A)^2} dA \cdot \frac{A}{A} = \frac{A}{(1+\beta A)} \cdot \frac{1}{(1+\beta A)} \cdot \frac{dA}{A} \quad \Big/ : A_f$$

$$\boxed{\frac{dA_f}{A_f} = \frac{1}{1+\beta A} \cdot \frac{dA}{A}}$$

za toliko je umanjeno

osjetljivost pojačanja uz primjenjive P.V.

osjetljivost pojačanja A grane

Primjer 6.1) PPV

$A_i = 10^4$ $\qquad \frac{dA}{A} = 20\%$

$\beta = 0{,}01$

$A_f = ?$

$$\frac{dA_f}{A_f} = \frac{1}{1+\beta A} \cdot \frac{dA}{A} = \frac{1}{1+\underbrace{0{,}01 \cdot 10^4}_{100}} \cdot \frac{dA}{A}$$

$$= \frac{1}{101} \cdot 20$$

$$\boxed{\frac{dA_f}{A_f} \approx 0{,}2\%}$$

Linearizacija prijenosne karakteristike pojačala

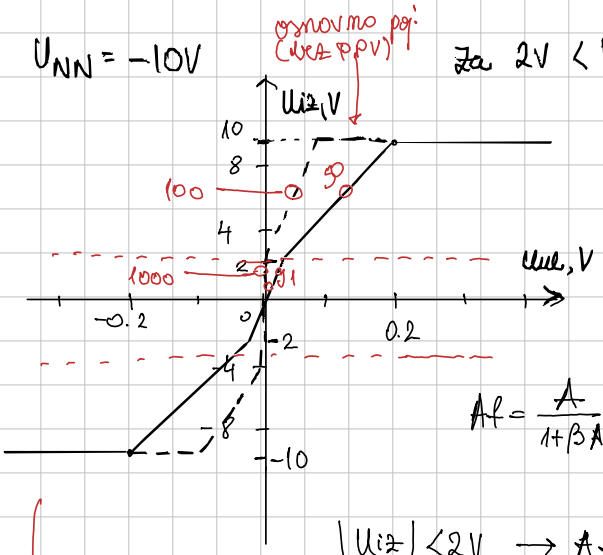
Primjer 6.2.) Da steknemo osjećaj kroz brojke

$U_{PP} = 10V$

$U_{NN} = -10V$

za $|u_{iz}| < 2 \longrightarrow A_{v_1} = 1000$ → puno veći nagib

za $2V < |u_{iz}| < 10V \longrightarrow A_{v_1} = 100$



prijenosna karakt $A_{vf} = ?$ PPV

$\beta = 0{,}01$

$A_f = \frac{A}{1+\beta A}$ → trebamo izračunati za 2 intervala

$|u_{iz}| < 2V \rightarrow A_f = \frac{1000}{1+0{,}01\cdot 1000} \Rightarrow \boxed{A_{f_1} = 90{,}9}$

$|u_{iz}| > 2V \rightarrow A_f = \frac{100}{1+0{,}01\cdot 100} \Rightarrow \boxed{A_{f_2} = 50}$

pojačalo s primjenjenom PV, nagib pojačala je ≈ 90 → puno blaži (puno manje pojačanje) od normalnog pojačala

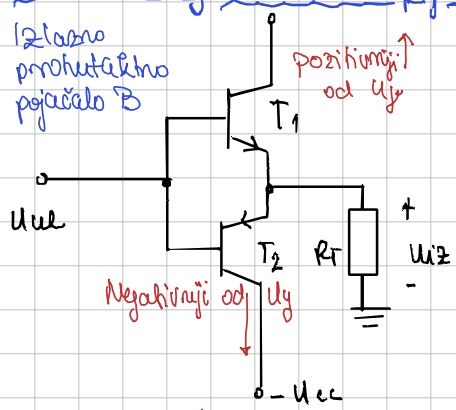
▶ uz PPV puno linearnija karakteristika

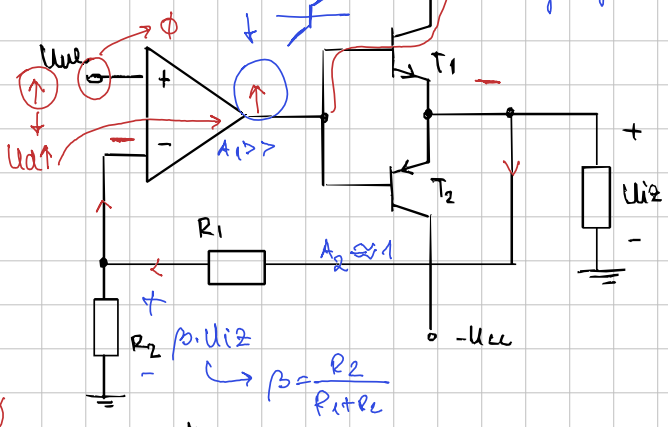
Linearizacija karakt pojačala snage:

A grama $A = A_1 \cdot A_2 \approx A_1$

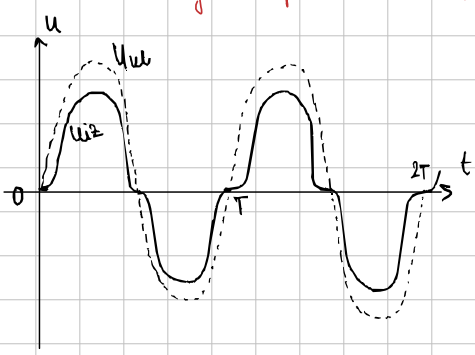
skok $\approx 0{,}7V$

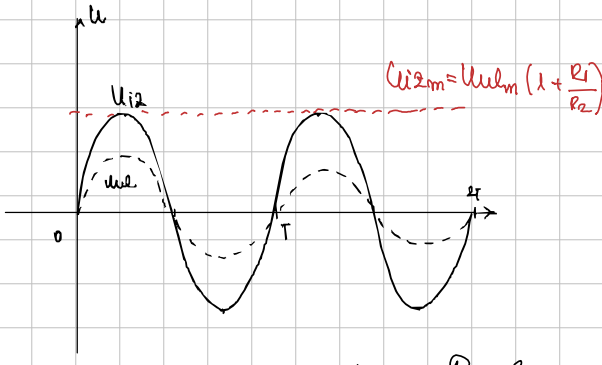
uključuje PV





▶ ima izobličenja kada ulazni signal prolazi kroz Ø





$u_{ul} \approx \beta \cdot u_{iz} \Longrightarrow \frac{u_{iz}}{u_{ul}} = \frac{1}{\beta} = \frac{R_1+R_2}{R_2}$

$\frac{u_{iz}}{u_{ul}} = 1 + \frac{R_1}{R_2}$

PROŠIRENJE FREKV. POJASA

~ Visoke frekv. ~

Frekv. odziv osnovnog pojačala i koeficijent PV

$$A(j\omega) = \frac{A_0}{1 + j\frac{\omega}{\omega_g}}$$

ω_g – granična frekv.

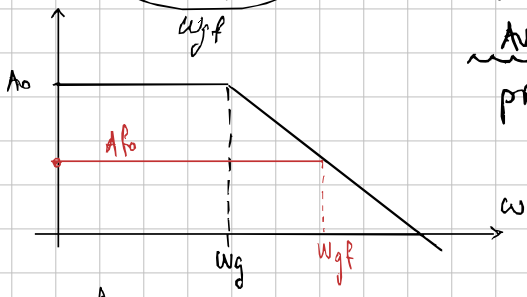
$\beta = \beta_0 \neq f(j\omega)$

$$A_f(j\omega) = \frac{A(j\omega)}{1 + \beta \cdot A(j\omega)} = \frac{\frac{A_0}{1 + j\frac{\omega}{\omega_g}}}{1 + \beta \frac{A_0}{1 + j\frac{\omega}{\omega_g}}} = \frac{\frac{A_0}{\left(\quad\right)}}{\frac{1 + j\frac{\omega}{\omega_g} + \beta A_0}{\left(\quad\right)}}$$

(faktorizirano frekv. ovisne članove)

$$A_{f0} = \frac{A_0}{1 + \beta A_0} \cdot \frac{1}{1 + j\frac{\omega}{\underbrace{\omega_g \cdot \frac{1}{1+\beta A_0}}_{\omega_{gf}}}} = A_{f0} \cdot \frac{1}{1 + j\frac{\omega}{\omega_{gf}}}$$

$$\boxed{\omega_{gf} = \omega_g(1 + \beta A_0)}$$



$A_0 \cdot \omega_g = A_{f0} \cdot \omega_{gf}$

približno konstantno

$$A_{f0} = \frac{A_0}{1 + \beta A_0}$$

$$\omega_{gf} = \omega_g(1 + \beta A_0)$$

$$A_{f0} \cdot \omega_{gf} = \frac{A_0}{\cancel{1 + \beta A_0}} \cdot \omega_g(\cancel{1 + \beta A_0}) \longrightarrow A_{f0} \cdot \omega_{gf} = A_0 \cdot \omega_g$$

Primjer: Moramo napraviti pojačalo parametara:

$A = 100$

$\omega_g = 100\,MHz$

korištenjem

$A = 100$

$\omega_g = 10\,MHz$

→ koristimo 2 O.P. i primjenili neg. PV.

$$\boxed{\begin{aligned} A_1 &= 10 \\ \omega_{gf_1} &= 100\,MHz \cdot 2\pi \end{aligned}}$$

$$A \cdot \omega_g = A_1 \cdot \omega_{gf_1}$$

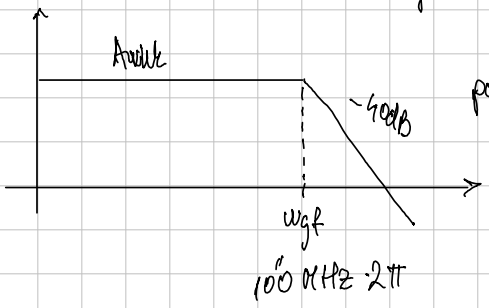
$\frac{1}{10}$

$A_2 = 10$

$\omega_{gf} = 100\,MHz \cdot 2\pi$

$$A_{uk} = A_1 \cdot A_2 \longrightarrow \underline{\underline{A_{uk} = 100}} \rightarrow \left(\frac{A_1}{1 + j\frac{\omega}{\omega_{gf_1}}}\right)\left(\frac{A}{1 + j\frac{\omega}{\omega_{gf_2}}}\right) = \frac{A_{uk}}{\left(1 + j\frac{\omega}{\omega_{gfuk}}\right)^2}$$

$\omega_{guk} \approx 100\,MHz \cdot 2\pi$



povećali smo širinu frekv. pojasa 10 x

Niske frekvencije

$$A(j\omega) = \frac{j\frac{\omega}{\omega_d}}{1+j\frac{\omega}{\omega_d}}$$

$$A(j\omega) = A_0 \cdot \frac{j\frac{\omega}{\omega_d}}{1+j\frac{\omega}{\omega_d}} = \frac{A_0 \frac{j\omega/\omega_d}{1+j\frac{\omega}{\omega_d}}}{1+\beta A_0 \cdot \frac{j\omega/\omega_d}{1+j\frac{\omega}{\omega_d}}} = \dots = \underbrace{\frac{A_0}{1+\beta A_0}}_{A_{f0}} \cdot \frac{j\frac{\omega}{\omega_d}}{j\frac{\omega}{\omega_d}+\frac{1}{1+\beta A_0}}$$

$$A(j\omega) = \frac{A_0}{1+\beta A_0} \cdot \frac{j\frac{\omega}{\omega_d}}{\frac{j\omega(1+\beta A_0)+\omega_d}{\omega_d(1+\beta A_0)}} = \underbrace{\frac{A_0}{1+\beta A_0}}_{A_{f0}} \cdot \frac{j\frac{\omega}{\omega_d}(1+\beta A_0)}{1+j\frac{\omega(1+\beta A_0)}{\omega_d}}$$

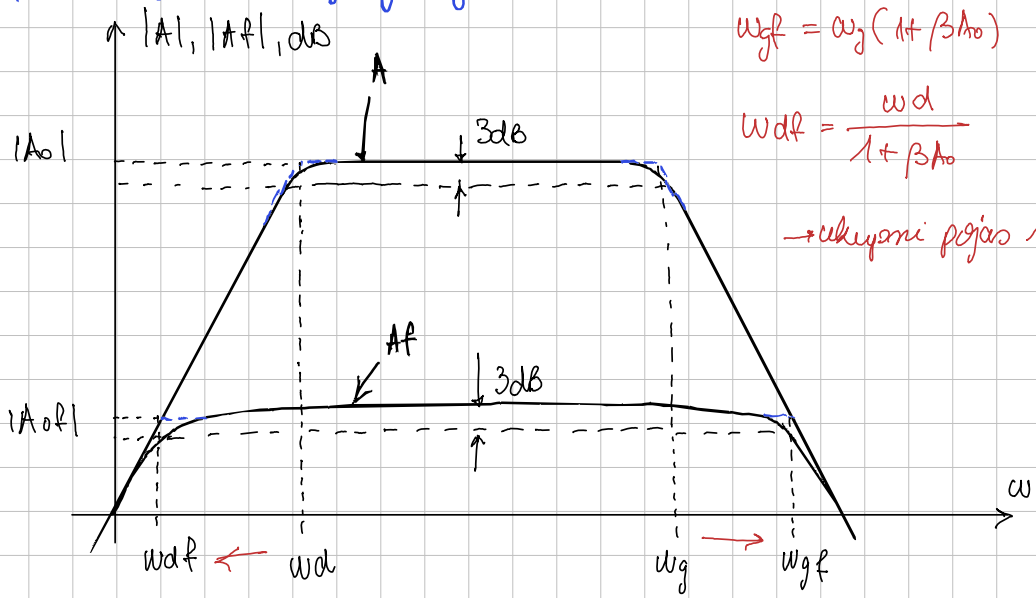
ω_{df}

donja granična uz primjenu PV

$$\boxed{A_f(j\omega) = A_{f0} \cdot \frac{j\frac{\omega}{\omega_{df}}}{1+j\frac{\omega}{\omega_{df}}}}$$

$$\omega_{df} = \frac{\omega_d}{1+\beta A_0} \; ; \; \omega_{df} \downarrow$$

Proširenje frekvencijskog pojasa:



$$\omega_{gf} = \omega_g(1+\beta A_0)$$

$$\omega_{df} = \frac{\omega_d}{1+\beta A_0}$$

→ukupni pojas se širi